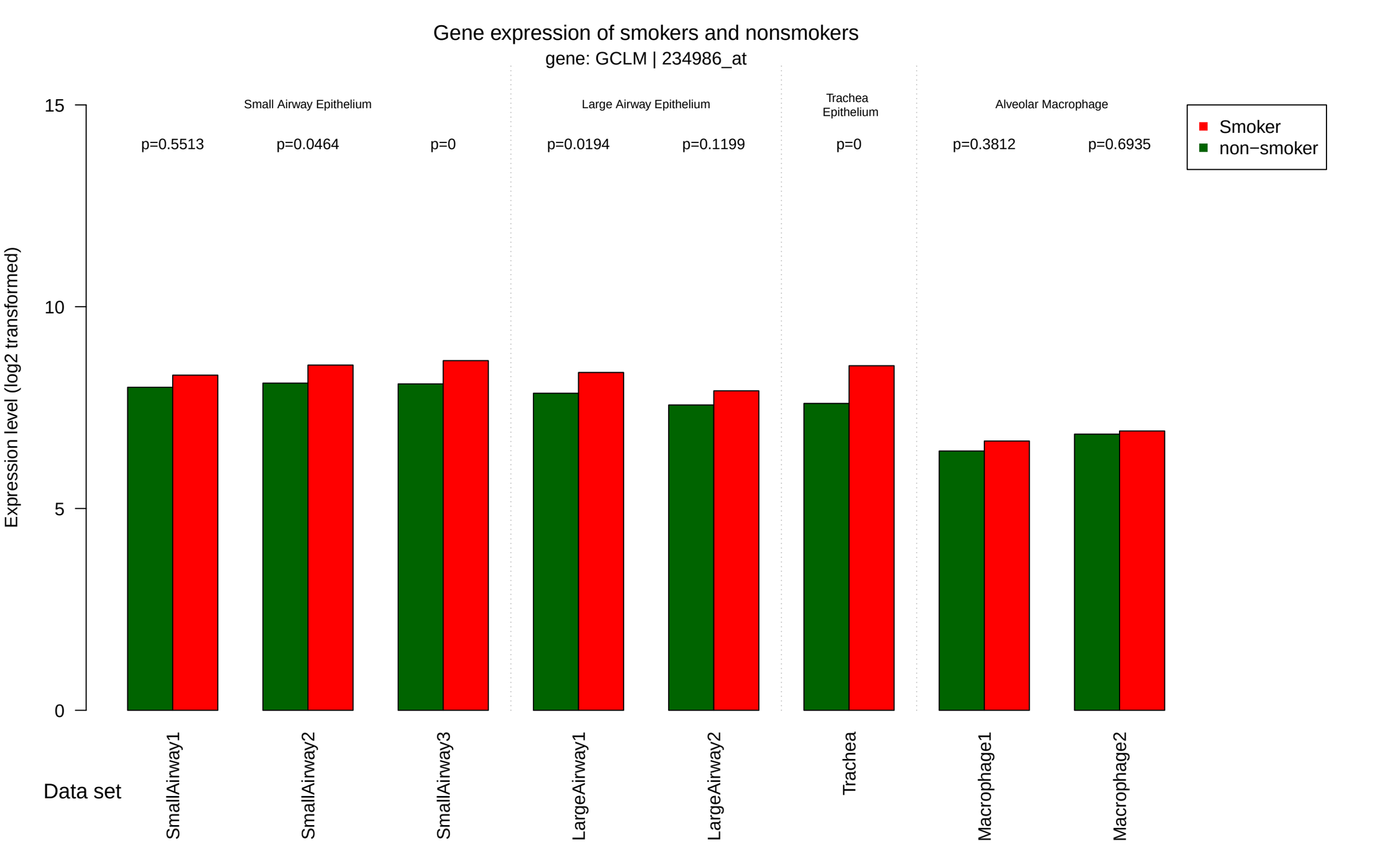
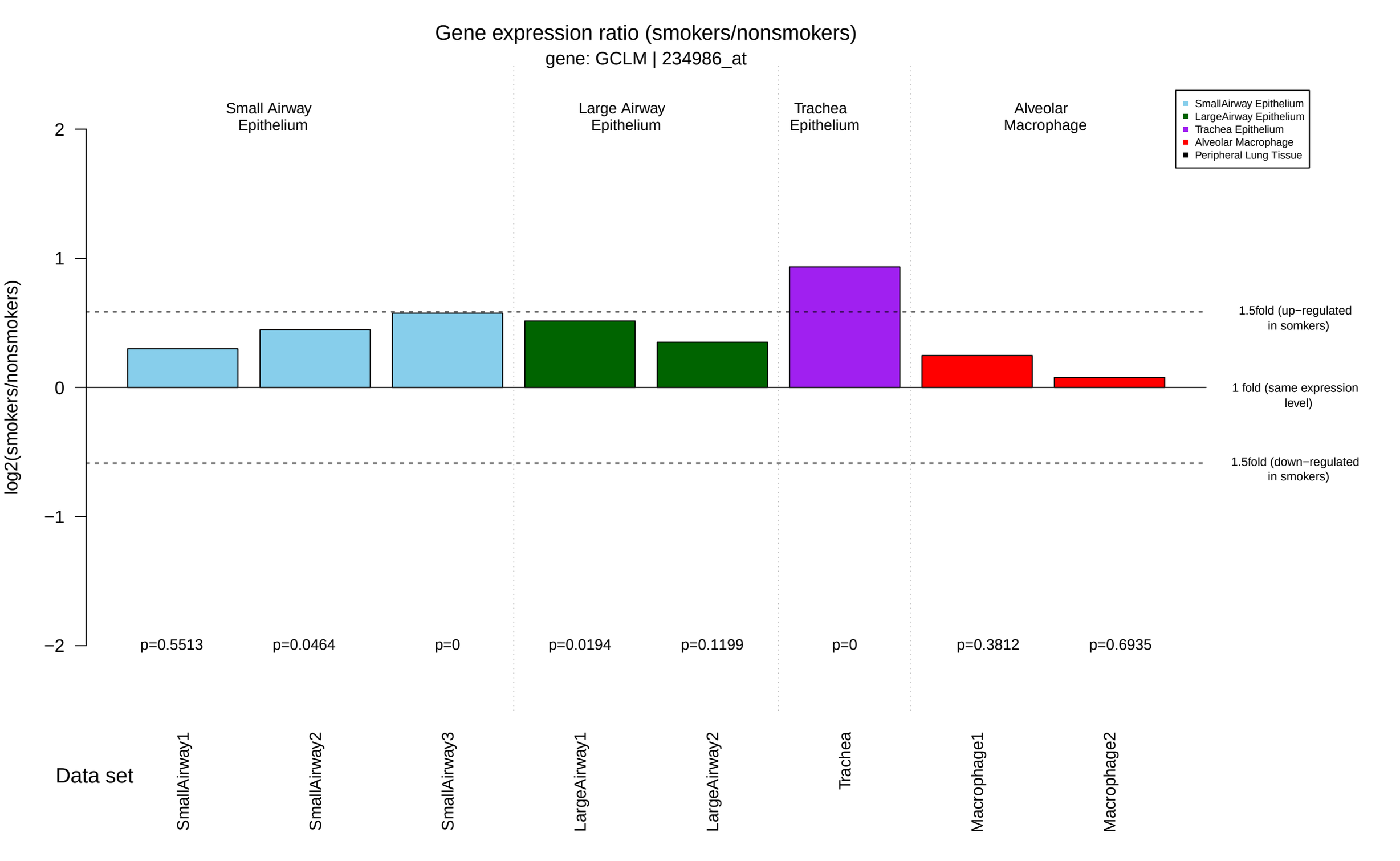
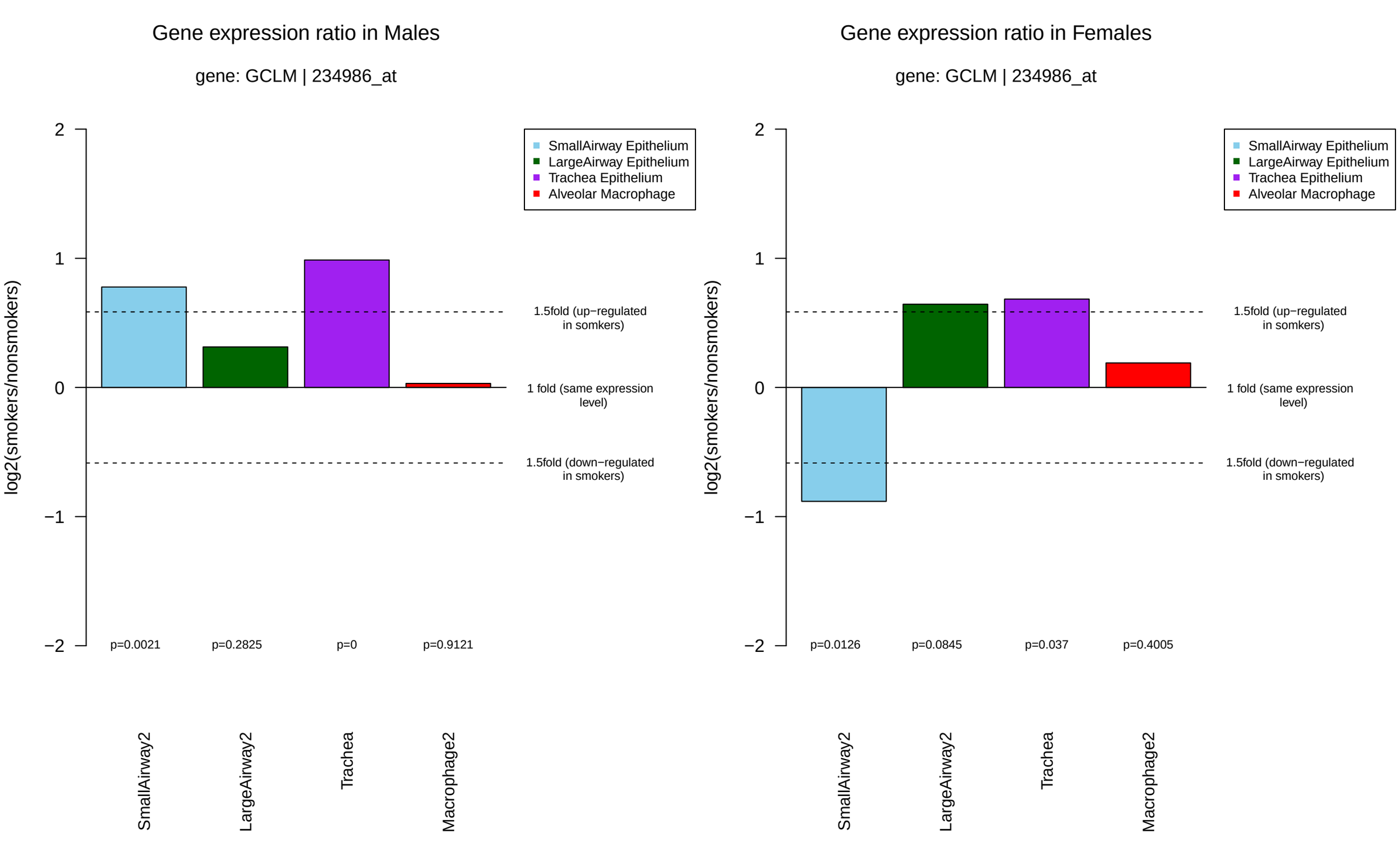
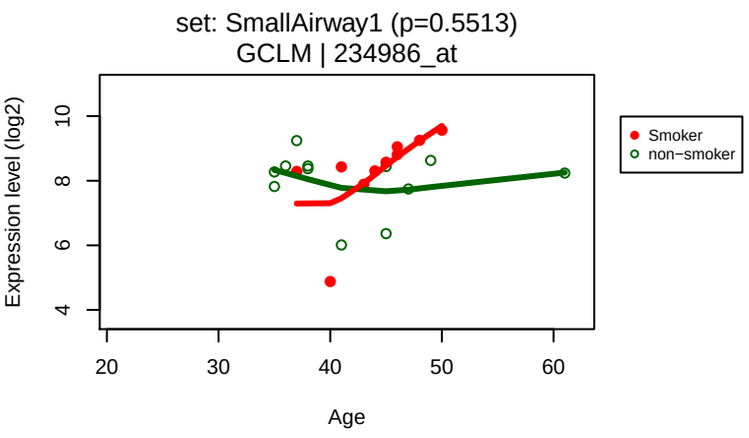
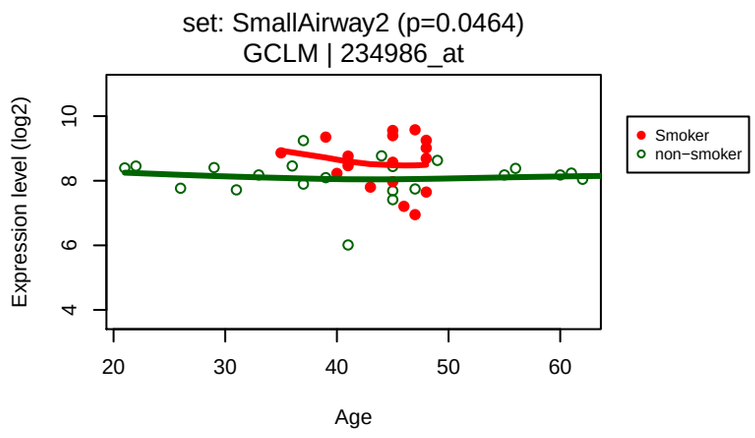
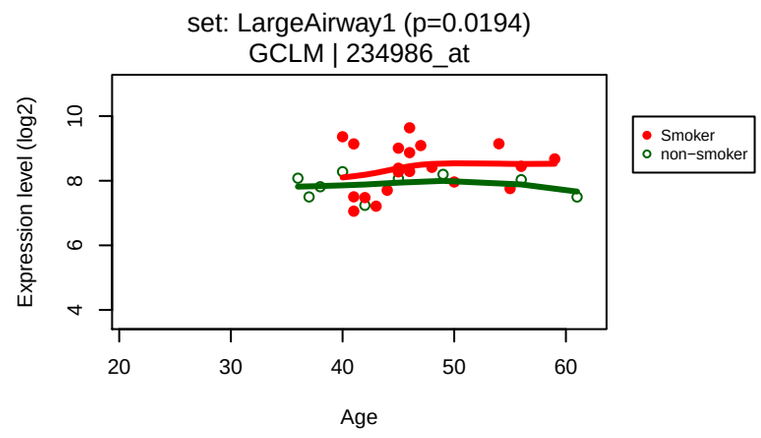

Gene expression of smokers and nonsmokers
gene: GCLM | 234986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5513
p=0.0464
p=0
p=0.0194
p=0.1199
p=0
p=0.3812
p=0.6935
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

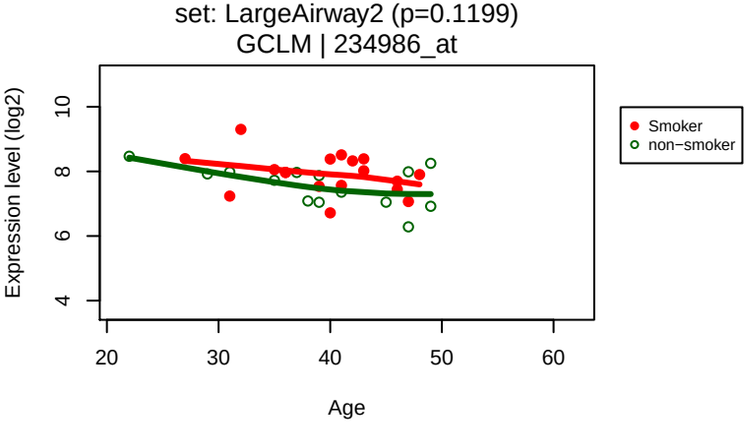

Gene expression ratio (smokers/nonsmokers)
gene: GCLM | 234986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5513
p=0.0464
p=0
p=0.0194
p=0.1199
p=0
p=0.3812
p=0.6935
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

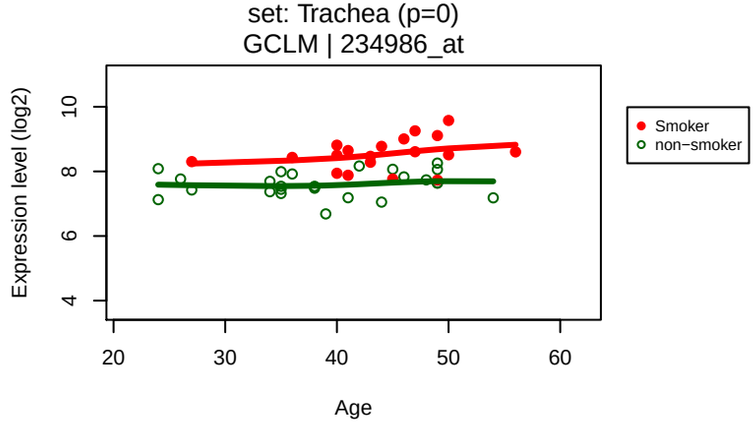

Gene expression ratio in Males
gene: GCLM | 234986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.2825
p=0
p=0.9121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCLM | 234986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0126
p=0.0845
p=0.037
p=0.4005
SmallAirway2
LargeAirway2
Trachea
Macrophage2

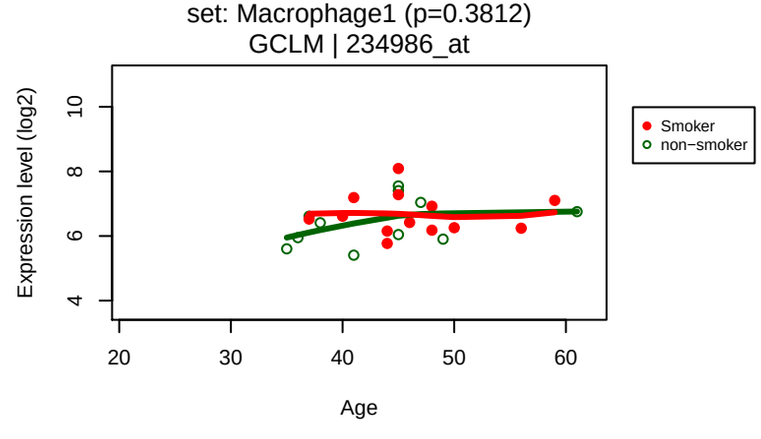

set: SmallAirway1 (p=0.5513)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker

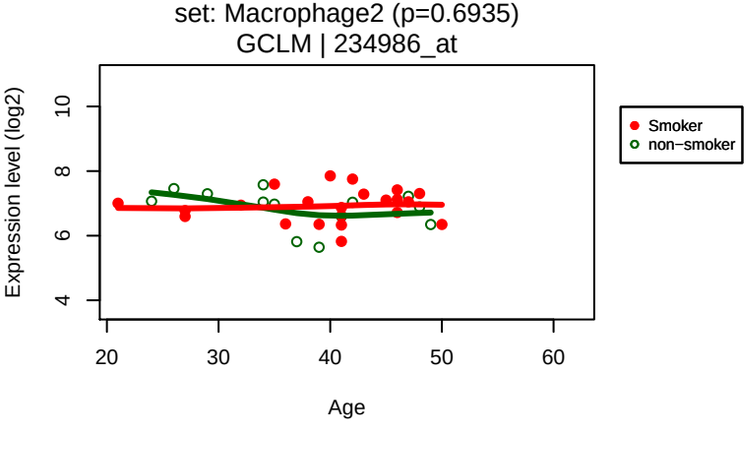

set: SmallAirway2 (p=0.0464)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker

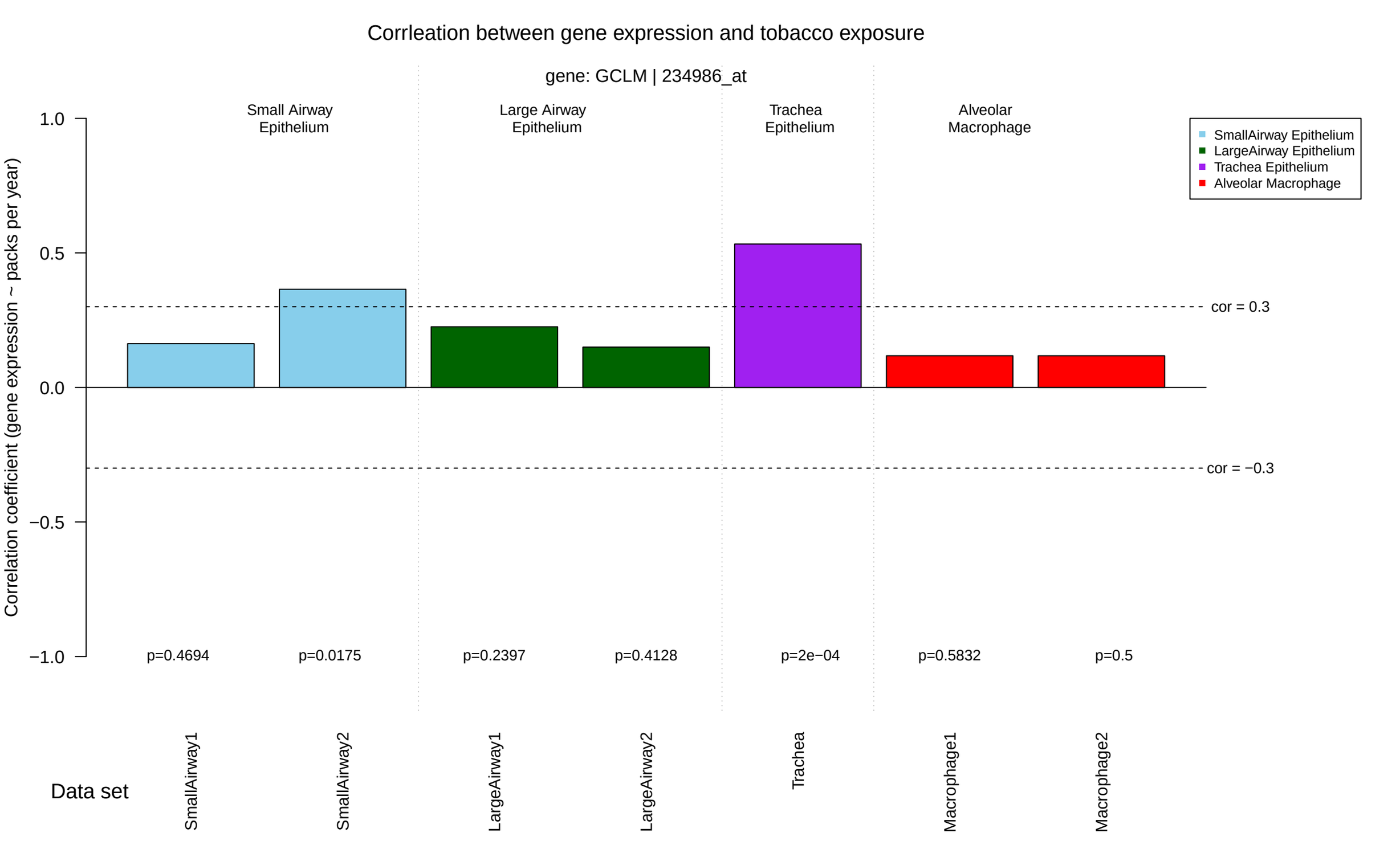

set: LargeAirway1 (p=0.0194)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1199)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3812)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6935)
GCLM | 234986_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCLM | 234986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4694
p=0.0175
p=0.2397
p=0.4128
p=2e−04
p=0.5832
p=0.5
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2